

Practice Examination - 2

Application Build and Release Automation Lab

Campus Health Appointment System

Student Name: Ajay Kumar

Roll Number: 22515990001

Program: B.Tech Computer Science and Engineering

Semester: 5th Semester

Subject Code: BCSE1522

Question 1 - Linux Project Setup

Project directory, folders, health_info.txt, pwd, ls, cat and tree.

```
View this pull request on GitHub: https://github.com/IshitaNainwal/Cloud-3rd-year-26-27/pull/21
ajaykumar@ajays-MacBook-Air Cloud-3rd-year-26-27 % gh pr view 21 --repo IshitaNainwal/Cloud-3rd-year-26-27 --json state,mergedAt
{
  "mergedAt": null,
  "state": "OPEN"
}
ajaykumar@ajays-MacBook-Air Cloud-3rd-year-26-27 % cd ~/Documents
ajaykumar@ajays-MacBook-Air Documents % pwd
/Users/ajaykumar/Documents
ajaykumar@ajays-MacBook-Air Documents % mkdir -p Ajay-GLA-HealthSystem
cd Ajay-GLA-HealthSystem
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % mkdir src docs config backup
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % cat > health_info.txt <<'EOF'
Student Name: Ajay Kumar
Roll Number: 22515990001
Subject Name: Application Build and Release Automation Lab
Project Name: Campus Health Appointment System
Semester: 5th Semester
EOF
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % pwd
/Users/ajaykumar/Documents/Ajay-GLA-HealthSystem
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % ls
backup      config      docs        health_info.txt  src
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % cat health_info.txt
Student Name: Ajay Kumar
Roll Number: 22515990001
Subject Name: Application Build and Release Automation Lab
Project Name: Campus Health Appointment System
Semester: 5th Semester
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % tree
.
├── backup
├── config
├── docs
├── health_info.txt
└── src

5 directories, 1 file
```

Question 2 - Git Version Control (Part 1)

Git initialization, identity, status, staging and initial commit.

```
5 directories, 1 file
[ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git init
Initialized empty Git repository in /Users/ajaykumar/Documents/Ajay-GLA-HealthSystem/.git/
[ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git branch -M main
[ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % touch src/.gitkeep docs/.gitkeep backup/.gitkeep
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git config --global user.name
git config --global user.email
Ajay Chaudhary
ajaysikarwar9636@gmail.com
[ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    backup/
    config/
    docs/
    health_info.txt
    src/

nothing added to commit but untracked files present (use "git add" to track)
[ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git add .
[ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   backup/.gitkeep
    new file:   config/.gitkeep
    new file:   docs/.gitkeep
    new file:   health_info.txt
    new file:   src/.gitkeep

[ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git commit -m "Initialize Campus Health Appointment System"
[main (root-commit) 9097efc] Initialize Campus Health Appointment System
 5 files changed, 5 insertions(+)
 create mode 100644 backup/.gitkeep
 create mode 100644 config/.gitkeep
 create mode 100644 docs/.gitkeep
 create mode 100644 health_info.txt
 create mode 100644 src/.gitkeep
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % cat >> health_info.txt <<'EOF'

Project Objective: To provide an easy system for students to book and manage campus health appointments.

Program/Course: B.Tech Computer Science and Engineering
EOF
[ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % cat health_info.txt
Student Name: Ajay Kumar
Roll Number: 22515990001
Subject Name: Application Build and Release Automation Lab
Project Name: Campus Health Appointment System
Semester: 5th Semester

Project Objective: To provide an easy system for students to book and manage campus health appointments.
```

Question 2 - Git Version Control (Part 2)

health_info.txt modification, git diff/status, second commit and history.

```
Program/Course: B.Tech Computer Science and Engineering
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git diff
diff --git a/health_info.txt b/health_info.txt
index a1b918d..209b221 100644
--- a/health_info.txt
+++ b/health_info.txt
@@ -3,3 +3,7 @@ Roll Number: 22515990001
 Subject Name: Application Build and Release Automation Lab
 Project Name: Campus Health Appointment System
 Semester: 5th Semester
+
+Project Objective: To provide an easy system for students to book and manage campus health appointments.
+
+Program/Course: B.Tech Computer Science and Engineering
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   health_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git add health_info.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git commit -m "Add project objective and student course"
[main 21a3b34] Add project objective and student course
1 file changed, 4 insertions(+)
```


Question 3 - Independent Feature Development

appointment-feature branch, appointment.txt, commit, branch/history verification and return to main.

```
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git add health_info.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git commit -m "Add project objective and student course"
[main 21a3b34] Add project objective and student course
1 file changed, 4 insertions(+)
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git log --oneline
21a3b34 (HEAD -> main) Add project objective and student course
9097efc Initialize Campus Health Appointment System
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git switch -c appointment-feature
Switched to a new branch 'appointment-feature'
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git branch --show-current
appointment-feature
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % cat > src/appointment.txt <<'EOF'
Appointment Module
```

Purpose:
This module allows students to book and manage campus health appointments easily.

Features:

1. Book a new health appointment.
2. View scheduled appointments.
3. Cancel an existing appointment.
4. Select preferred appointment date and time.

Expected Users:
Students and campus health staff.

```
EOF
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % cat src/appointment.txt
Appointment Module
```

Purpose:
This module allows students to book and manage campus health appointments easily.

Features:

1. Book a new health appointment.
2. View scheduled appointments.
3. Cancel an existing appointment.
4. Select preferred appointment date and time.

Expected Users:
Students and campus health staff.

```
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git add src/appointment.txt
git commit -m "Develop appointment module"
[appointment-feature 7da0399] Develop appointment module
1 file changed, 13 insertions(+)
create mode 100644 src/appointment.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git branch
* appointment-feature
main
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git branch --show-current
appointment-feature
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git log --oneline --decorate --all
7da0399 (HEAD -> appointment-feature) Develop appointment module
21a3b34 (main) Add project objective and student course
9097efc Initialize Campus Health Appointment System
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git switch main
Switched to branch 'main'
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git branch --show-current
main
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % ls src
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % █
```

Question 4 - Merge and Resolve Conflict (Part 1)

Feature merge, system-update branch, parallel project-name edits and merge conflict creation.

```
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git merge appointment-feature
Updating 21a3b34..7da0399
Fast-forward
 src/appointment.txt | 13 ++++++++
 1 file changed, 13 insertions(+)
 create mode 100644 src/appointment.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % ls src
appointment.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % cat src/appointment.txt
Appointment Module

Purpose:
This module allows students to book and manage campus health appointments easily.

Features:
1. Book a new health appointment.
2. View scheduled appointments.
3. Cancel an existing appointment.
4. Select preferred appointment date and time.

Expected Users:
Students and campus health staff.
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git switch -c system-update
Switched to a new branch 'system-update'
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % awk '{if ($0=="Project Name: Campus Health Appointment System") print "Project Name: GLA Health Appointment Portal"; else print}' health_info.txt > temp.txt
t && mv temp.txt health_info.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % grep "Project Name" health_info.txt
Project Name: GLA Health Appointment Portal
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git add health_info.txt
git commit -m "Update project name in system-update branch"
[system-update 6a566e7] Update project name in system-update branch
 1 file changed, 1 insertion(+), 1 deletion(-)
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git switch main
Switched to branch 'main'
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % awk '{if ($0=="Project Name: Campus Health Appointment System") print "Project Name: GLA Campus Medical Booking System"; else print}' health_info.txt > temp.txt
p.txt && mv temp.txt health_info.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % grep "Project Name" health_info.txt
Project Name: GLA Campus Medical Booking System
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git add health_info.txt
git commit -m "Update project name on main branch"
[main c07b67e] Update project name on main branch
 1 file changed, 1 insertion(+), 1 deletion(-)
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git merge system-update
Auto-merging health_info.txt
CONFLICT (content): Merge conflict in health_info.txt
Automatic merge failed; fix conflicts and then commit the result.
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git status
cat health_info.txt
On branch main
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
    both modified:   health_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
Student Name: Ajay Kumar
Roll Number: 22515990001
```


Question 4 - Merge and Resolve Conflict (Part 2)

Conflict markers, manual resolution, staging, merge commit and final history.

```
no changes added to commit (use "git add" and/or "git commit -a")
Student Name: Ajay Kumar
Roll Number: 22515990001
Subject Name: Application Build and Release Automation Lab
<<<<<< HEAD
Project Name: GLA Campus Medical Booking System
=====
Project Name: GLA Health Appointment Portal
>>>>>> system-update
Semester: 5th Semester

Project Objective: To provide an easy system for students to book and manage campus health appointments.

Program/Course: B.Tech Computer Science and Engineering
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % cat > health_info.txt <<'EOF'
Student Name: Ajay Kumar
Roll Number: 22515990001
Subject Name: Application Build and Release Automation Lab
Project Name: GLA Campus Health Appointment System
Semester: 5th Semester

Project Objective: To provide an easy system for students to book and manage campus health appointments.

Program/Course: B.Tech Computer Science and Engineering
|EOF
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % cat health_info.txt
Student Name: Ajay Kumar
Roll Number: 22515990001
Subject Name: Application Build and Release Automation Lab
Project Name: GLA Campus Health Appointment System
Semester: 5th Semester

Project Objective: To provide an easy system for students to book and manage campus health appointments.

Program/Course: B.Tech Computer Science and Engineering
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % grep -n '<<<<<<\\|=====\\|>>>>>> ' health_info.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git add health_info.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git status
On branch main
All conflicts fixed but you are still merging.
(use "git commit" to conclude merge)

Changes to be committed:
  modified:   health_info.txt

ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git commit -m "Resolve project name merge conflict"
[main ff760dd] Resolve project name merge conflict
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git status
On branch main
nothing to commit, working tree clean
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git log --oneline --graph --decorate --all
*   ff760dd (HEAD -> main) Resolve project name merge conflict
|
| * 6a566e7 (system-update) Update project name in system-update branch
| * | c07b67e Update project name on main branch
|/
* 7da0399 (appointment-feature) Develop appointment module
* 21a3b34 Add project objective and student course
* 9097efc Initialize Campus Health Appointment System
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % █
```

Question 5 - Repository Demonstration

Final project structure, clean status, branches, commit graph, appointment module and health_info.txt.

```
| \
| * 6a566e7 (system-update) Update project name in system-update branch
* | c07b67e Update project name on main branch
| /
* 7da0399 (appointment-feature) Develop appointment module
* 21a3b34 Add project objective and student course
* 9097efc Initialize Campus Health Appointment System
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % tree
.
├── backup
├── config
├── docs
├── health_info.txt
├── src
└── appointment.txt

5 directories, 2 files
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git status
On branch main
nothing to commit, working tree clean
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git branch
* main
  system-update
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % git log --oneline --graph --decorate --all
* ff760dd (HEAD -> main) Resolve project name merge conflict
| \
| * 6a566e7 (system-update) Update project name in system-update branch
* | c07b67e Update project name on main branch
| /
* 7da0399 (appointment-feature) Develop appointment module
* 21a3b34 Add project objective and student course
* 9097efc Initialize Campus Health Appointment System
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % ls src
appointment.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % cat src/appointment.txt
Appointment Module

Purpose:
This module allows students to book and manage campus health appointments easily.

Features:
1. Book a new health appointment.
2. View scheduled appointments.
3. Cancel an existing appointment.
4. Select preferred appointment date and time.

Expected Users:
Students and campus health staff.
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % cat health_info.txt
Student Name: Ajay Kumar
Roll Number: 22515990001
Subject Name: Application Build and Release Automation Lab
Project Name: GLA Campus Health Appointment System
Semester: 5th Semester

Project Objective: To provide an easy system for students to book and manage campus health appointments.

Program/Course: B.Tech Computer Science and Engineering
ajaykumar@ajays-MacBook-Air Ajay-GLA-HealthSystem % █
```